

Proof of Concept for Spectrum Sharing between Terrestrial and Satellite Networks

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Abstract—Satellite communication systems are fundamental components to deploy the future smart and sustainable networks and to achieve the ambitious goal of bringing wireless connectivity anywhere, anytime, at any device. In this new role, one of the main challenges that satellite communication component has to face is the maximization of the spectrum usage. 3GPP communication technologies are extended from Terrestrial Networks (TNs) to Non-Terrestrial Networks (NTNs), but so far the standardisation efforts have been focused on systems where TNs and NTNs operate in their dedicated frequency bands.

In this paper, a dynamic spectrum sharing model between NTN elements, in a Non Geostationary orbit, and TN is proposed. A Proof of Concept (PoC) is developed, in order to carry out the interference protection computation. We show that the developed spectrum sharing model can enable spectrum sharing between NTN and TN when their coverage areas do not have to overlap, that the sharing arrangement increases significantly the service coverage in the frequency band and slightly improves the spectrum utilization efficiency. It is also shown that the spectrum management system is able to manage the interference level and to keep the interference-to-noise ratio at the TN user equipment below the specified limit. In fact, the aggregate interference caused by the sharing arrangement does not decrease the capacity of the TN downlink.

I. INTRODUCTION

The satellite access to 5G and 6G services have the ambition to literally extend the coverage of the terrestrial networks to everywhere on Earth. The unlimited coverage brings the Internet services to people who have been unconnected so far due to lack of coverage and ensures service continuity. Many countries build their critical communication network for public protection and disaster recovery based on the same infrastructure and technology as the commercial cellular networks. The unlimited coverage is a prerequisite for such operations. While dedicating a band for just 5G or 6G satellite communication is technically possible, there are two main reasons why the spectrum sharing has to be studied carefully: it is extremely difficult to find globally available and wide enough spectrum band; through sharing the same frequency bands with TNs, the selection and cost level of UEs for satellite communication can be strongly improved.

Dynamic Spectrum Access (DSA) controls the use of radio spectrum resources so that spectrum users do not cause harmful interference to each other. ITU-R Radio Regulations [1] define harmful interference as interference which endangers the functioning of a radionavigation service or of other safety services or seriously degrades, obstructs, or repeatedly interrupts a radiocommunication service operating in accordance with Radio Regulations. DSA can also optimize the efficient use of radio spectrum resources or maintain a fair allocation between spectrum users, but often such a capability is called co-existence management. Radio spectrum management, which also avoids harmful interference, becomes dynamic when the rate of changes is faster than the license handling process of a National Radio Administration. When the radio resources are managed on a physical and Medium Access Control (MAC) layer, and especially when the radio resources are managed within the same operator's networks, the spectrum management is rather referred as scheduling, Radio Resource Control, Radio Resource Management or coordination [2]. When the spectrum management is dynamic, centralized, and the spectrum users are of different types, the system can in most cases be called DSA system with the exception of spectrum sharing between 4G and 5G systems, where the arrangement is named Dynamic Spectrum Sharing [3]. The following concepts are considered DSA: Television White Space [4], [5], Licensed Shared Access [6], [7], Citizens Broadband Radio Service [8], Tiered Spectrum Sharing Model [9] and Wi-Fi 6 Automated Frequency Coordination (AFC) [10]. They all apply a geolocation database for controlling the access to spectrum [11]–[13].

The sharing scenarios for satellite communications can be divided into four main categories: (1) secondary use of satellite spectrum by terrestrial systems [14]; (2) satellite systems as a secondary user of spectrum [15]; (3) extension of terrestrial network using satellite network [16]; and (4) two satellite systems sharing at least part of the same spectrum [17]. In the first two categories, spectrum sharing is based on static or dynamic spectrum sharing, in which users are categorised as Primary Users (PU) and Secondary Users (SU) of the spectrum. In two

other categories, spectrum sharing is coordinated between co-primary users. A comprehensive analysis of the four categories is provided in [18].

This paper describes a Proof of Concept (PoC) implementation of a spectrum management system for DSA to support Non-Geo-Stationary Orbits (NGSO) satellite 5G and 6G communication. The DSA system, studied here, assumes that the wireless systems sharing the spectrum are not compatible with each other, i.e., the sharing wireless networks are considered to be white noise to each other. We have divided DSA in cooperative and non-cooperative sharing. In cooperative sharing, the network operators have a common business interest, for example through common ownership, and therefore they are expected to tolerate a limited amount of interference from each other. In non-cooperative sharing, the allowed interference is defined by ITU-R and local regulative authority [19]. In the considered sharing scenarios, we assume that the NTN use is allowed if it does not cause harmful interference to TNs. With this PoC, we test the hypothesis that we can introduce NTN communication in the same frequency band as TN so that NTN does not cause harmful interference to TN and the geographical separation between TN and NTN is minimized. The PoC is implemented as a software tool as experimenting the control of real satellite transmissions would be extremely costly and difficult to arrange. To the authors' knowledge, this is the first time that a DSA system, which is able to activate and deactivate LEO satellite beams to protect TN deployments, is demonstrated as a proof of concept.

The rest of the paper is organized as follows: the spectrum sharing model is introduced in Section II, the Proof of Concept is described in Section III, the results of the experiment can be found in Section IV, and finally conclusions and future works are discussed in Section V.

II. SPECTRUM SHARING MODEL

The objective of the PoC discussed in this paper is to confirm the hypothesis that DSA management can be deployed for spectrum sharing between NGSO NTN and a TN in the same frequency band so that the network coverage in the band is significantly increased, while none or minimal performance degradation is caused to the primary TN. The PoC visualizes how spectrum sharing works and presents the benefits of both non-cooperative and cooperative DSA. The PoC consists of two parts: the spectrum management system, which computes the interference and controls the transmitters, and a visualization of the PoC. The purpose of the visualization is to select a sharing scenario, illustrate the sharing arrangement geographically, and present the Key Performance Indicators (KPIs) in the selected scenario.

Spectrum sharing presents several benefits. In particular, the primary system can operate without changes and knowledge of the DSA system, while the spectrum management system can protect primary use from harmful interference. DSA can adjust the interference according to the sharing agreement by adapting the secondary use to the changes in the primary

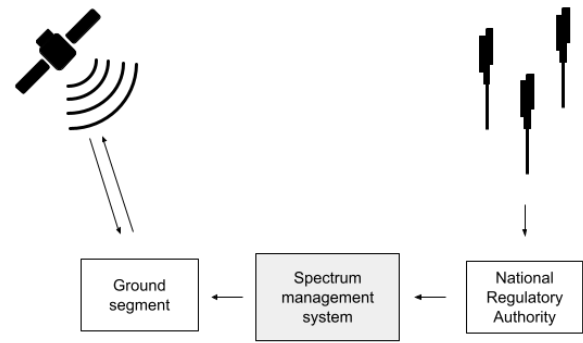


Figure 1: Spectrum management system as part of TN - NTN spectrum sharing.

use. In summary, the sharing arrangement does not cause practically any performance degradation.

The spectrum management system retrieves the TN sites and operational parameters from the National Regulatory Authorities. Based on the information about the satellite constellation and satellite antenna beams as a function of time, the spectrum management system computes the active beams at discrete moments of time so that the Interference to Noise ratio (I/N) at the TN user equipment does not exceed the set limit (-6 dB or 0 dB in this study for non-cooperative and cooperative sharing, respectively). Due to the computational complexity and to the amount of time required to compute the active beams, the computation is carried out offline, and the results are provided to the ground segment, see Figure 1. In the PoC, the active and inactive satellite beams are identified based on the aggregate interference caused to TN UEs. On a high level, the process works as follows

- 1) compute which UEs are TN UEs
- 2) perform linear optimization of the inactive and active beams
- 3) compute aggregate NTN signal level at TN UEs
- 4) compute KPIs: Interference, Capacity, Coverage, Spectrum Utilization Efficiency (SUE) and Minimum Separation Distance (MSD).

We use the 3GPP 38.811 propagation model [20] from space to earth, and ITU-R P.452 [21] for terrestrial signal paths. The antenna patterns for TN and NTN UEs and for terrestrial gNB are according to ITU-R F.1336 [22]. The satellite gNB antenna pattern is modified from ITU-R M.2101 [23] as illustrated in Figure 2. For selecting which UEs associate to TN gNBs, we create a constant grid of UEs in the study area. We compute the received power at UE from the closest gNBs by adding gNB transmit power (parameter), gNB antenna gain towards UE (F.1336), negative attenuation (P.452), and UE antenna gain (F.1336). If the highest power level is -106 dBm (parameter) or higher, the UE is considered as a TN UE.

By plotting the TN and NTN UEs with different colors on a map, we can show the coverage of TN and NTN. The KPIs can

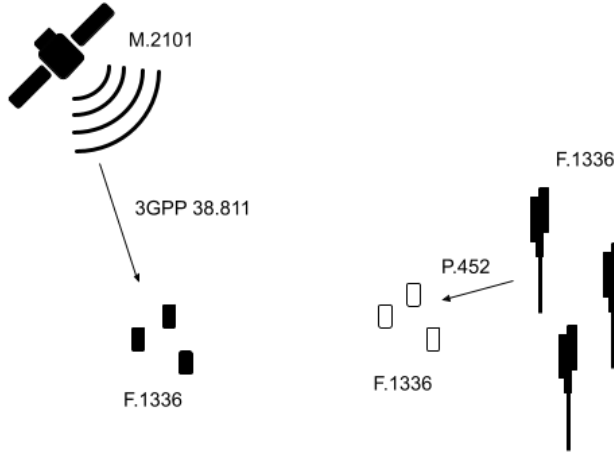


Figure 2: Propagation model and antenna pattern references

be computed after the association of the UEs to TN and NTN gNBs. Interference is expressed in terms of I/N , as it is done for the constraint functions in linear optimization procedure. Capacity is computed using an attenuated and truncated form of the Shannon bound [24]. SUE is defined as transferred bits (capacity) divided by area (number of UEs), time (1 second) and bandwidth. The absolute values of the SUE are not very informative, but when we compare the SUE of the no sharing and SUE without spectrum management with non-cooperative and cooperative spectrum sharing arrangements, we find out the gain achievable with the proposed DSA system. MSD is computed as the minimum distance between a TN UE and NTN UE.

III. PROOF OF CONCEPT

We consider a scenario where the TN has deployments on the ground, and the NTN gets permission to share the spectrum with the restriction that it is not allowed to cause harmful interference to TN. We define a geographic study area that is fully covered by NTN, which consists of two satellites with 91 beams each. The NTN beams centers are depicted in Figure 3. The satellite beams positions are from a LEO satellite constellation, which is partly illustrated in Figure 4 and globally in Figure 5. The TN is represented by Orange 4G sites in Centre-Val de Loire and in the surrounding French regions. The site locations have been downloaded from [25]. An example of Orange 4G sites in Centre-Val de Loire are presented in Figure 6. The regions which are partially covered by the study are: Centre-Val de Loire (Middle), Pays de la Loire (West), Normandie (North-West), Île de France (North), Grand Est (North-East), Bourgogne-Franche-Comté (East), Auvergne-Rhône-Alpes (South-East) and Nouvelle Aquitaine (South-West).

The decision if NTN causes harmful interference to TN is based on the I/N value at TN UE. The UE locations are represented as a constant grid in the study area. Interference is computed as aggregate interference from NTN beams only,

i.e., interference from other TN gNBs, other UEs or adjacent band interference are not taken into account. The beams are activated or disabled so that the maximum allowed I/N does not exceed the set limit in any of the TN UEs. In particular, the NTN beams can be active only if the aggregate interference to the TN UEs is below -6 dB in non-cooperative sharing and 0 dB in cooperative spectrum sharing. The above procedure is achieved by solving a linear optimization problem. The objective function is to maximize the number of active beams. The constraint functions ensure that the aggregated (sum) of received power from all NTN beams does not exceed the defined I/N limit for any TN UE. Each TN UE corresponds to a constraint function where the variable coefficient, of which the constraint matrix is formed, is computed by adding NTN gNB transmit power, which is considered a parameter, the NTN gNB antenna gain towards UE (M.2101), the negative attenuation (38.811), and the UE antenna gain (F.1336). For attenuation, the model takes into account the angle and distance between the NTN gNB and TN UE and clutter type from the clutter data. We run python linear optimization package with the objective and constraint function to find out which beams should be active and inactive. The decision which (non-TN) UEs are NTN UEs is made by computing the received power at UE from the closest NTN gNBs by adding gNB transmit power (parameter), gNB antenna gain towards UE (M.2101), negative attenuation (38.811), and UE antenna gain (F.1336). If the highest power level is -106 dBm (parameter) or higher, the UE is considered as an NTN UE. The optimization function is

$$\text{maximize } f(x_1, x_2, \dots, x_n) = x_1 + x_2 + \dots + x_n, x_n \in \{0, 1\} \quad (1)$$

where x_i is the state variable of the beam i . If the beam is active, it has value 1, if the beam is inactive, it has value 0. The constraint functions are

$$\begin{aligned} a_{11}x_1 + a_{21}x_2 + \dots + a_{n1}x_n &\leq b \\ a_{12}x_1 + a_{22}x_2 + \dots + a_{n2}x_n &\leq b \\ &\vdots \\ a_{1m}x_1 + a_{2m}x_2 + \dots + a_{nm}x_n &\leq b \end{aligned} \quad (2)$$

where a_{ij} is the field strength of beam i at UE j , and b is the maximum allowed field strength in the pixel. Parameter b is chosen slightly lower than the allowed power. The technical parameters used in the demonstration can be changed through the demonstration User Interface. A few configurable parameters and their default values can be found in Table I.

IV. EXPERIMENT RESULTS

In this PoC, we demonstrate how DSA can help satellite operators to get access in the frequency bands that are suitable for providing 5G, B5G, or 6G broadband connectivity. This is partially possible as many terrestrial capacity bands are only deployed in urban areas. The proposed DSA sharing arrangement allows satellite network deployments when they do not cause harmful interference to TNs. Without sharing,

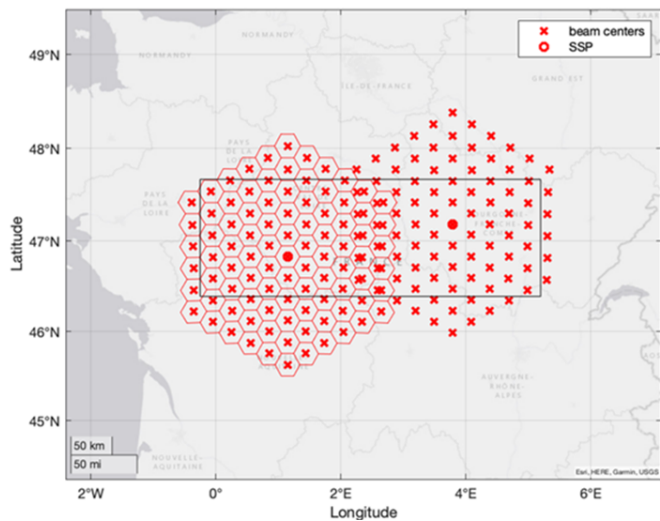


Figure 3: Satellite beams of two satellites cover the study area

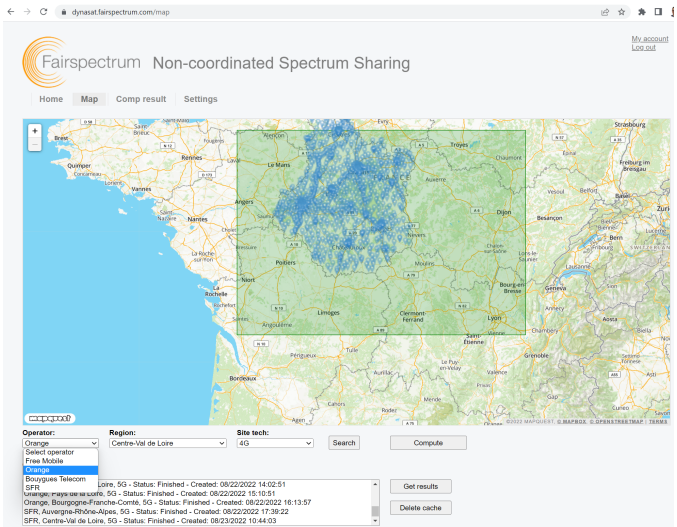


Figure 6: Study area and Orange 4G eNBs in Centre-Val de Loire, data from [25].



Figure 4: Studied satellite positions as part of the constellation over France. Source: Thales Alenia Space.

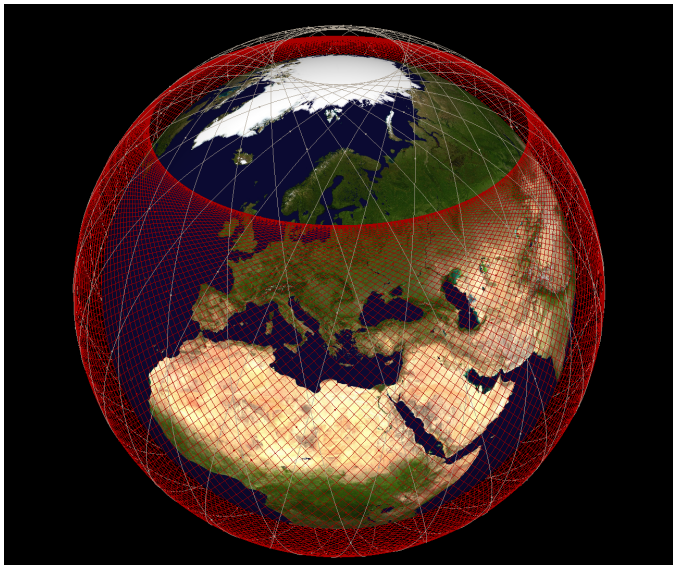


Figure 5: LEO satellite constellation example. Source: Thales Alenia Space.

Parameter	Value	Unit
Bandwidth	15.0	MHz
Coverage min SINR	0.0	dB
Coverage min rx power	-106.0	dBm
Frequency	2.4	GHz
TN gNB TX power	20.0	dBm
NTN gNB TX power	43.0	dBm

Table I: Proof of Concept configurable parameters and corresponding default values.

only TN coverage is present. In -6 dB I/N sharing, NTN can fill coverage gaps. The NTN coverage can further be enhanced when the TN and NTN operators cooperate by allowing a higher interference to TN which is presented in 0 dB I/N sharing in Figure 7.

The benefit of TN-NTN spectrum sharing is that NTN could access some of the International Mobile Telecommunications bands which are allocated to 3GPP terrestrial technologies. Sharing would simplify finding and allocating frequency bands for broadband NTN. The areas, which are not covered by TN, can be covered by NTN as illustrated in Figure 8. In the Proof of Concept example shown in Figure 7, without sharing, TN covers 22.94% of the study area. With $I/N -6$ dB sharing the TN coverage is 22.92%, and with $I/N 0$ dB sharing it is 22.79 %. So, there is a slight amount of harmful interference in terms of coverage caused by the sharing arrangement. Without interference management, the TN coverage would drop to 19.56%. NTN increases coverage in $I/N -6$ dB and 0 dB sharing by 41.61% and 43.79%, respectively.

Spectrum is a scarce and valuable resource. By sharing, SUE increases. When the number of TN gNBs is high compared with the beam number, the SUE improvement is present but small, see Figure 9. In the PoC example shown in Figure 7, compared with no sharing, SUE improves in $I/N -6$ dB and 0 dB sharing by 0.88% and 0.93%, respectively. Without

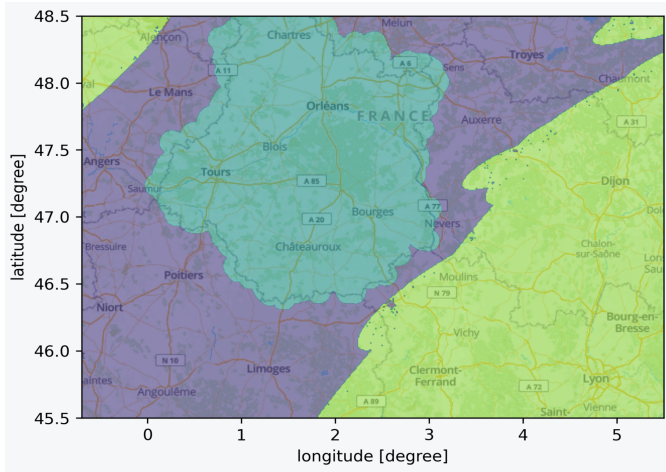


Figure 7: Illustration of I/N 0 dB sharing in Centre-Val de Loire, dark green TN, light green NTN, purple no coverage

interference management, SUE would drop by 41% compared to no sharing.

In the Proof of Concept example shown in Figure 7, I/N -6 dB and 0 dB curves have the maximum values of -7.23 dB and -1.60 dB, respectively. The maximum values are below the thresholds, confirming that the algorithm is working correctly. The spectrum management system can control the interference level at TN as depicted in Figure 10. Without interference management, I/N would vary between 20.7 dB and 25.4 dB.

All these benefits can be achieved without any or significant performance loss in TN. As no sharing (blue), -6 dB I/N (green) and 0 dB I/N (yellow) curves are very close to each other, we can conclude that this managed sharing does not decrease the capacity of the Terrestrial Network. No limit curve drops the median capacity per gNB from 89.4 Mbps to 43.5 Mbps in Figure 11. The capacity Cumulative Distribution Function (CDF) is computed per basestation. The maximum bandwidth (15 MHz) is equally divided between the UEs associated to the basestation. For each link, the capacity is computed according to 3GPP TR 38.803 section 5.2.7 [24] extended Shannon formula, which is a function of Signal to Noise Ratio (SNR) at UE. The bitrates of the downlinks to the associated UEs are summed to create the capacity value of the basestation.

It is worth noticing that cooperative sharing has a better coverage and a smaller MSD than non-cooperative sharing, while not causing any significant performance degradation.

V. CONCLUSIONS AND FUTURE WORK

In this paper, we considered spectrum sharing between terrestrial and non-terrestrial 3GPP networks. In developed countries, improving the TN coverage with NTN increases safety and lowers inequality between densely and scarcely populated areas. In developing countries, NTN provides a connectivity solution, which requires least maintenance and investments in infrastructure like site towers, site protection,

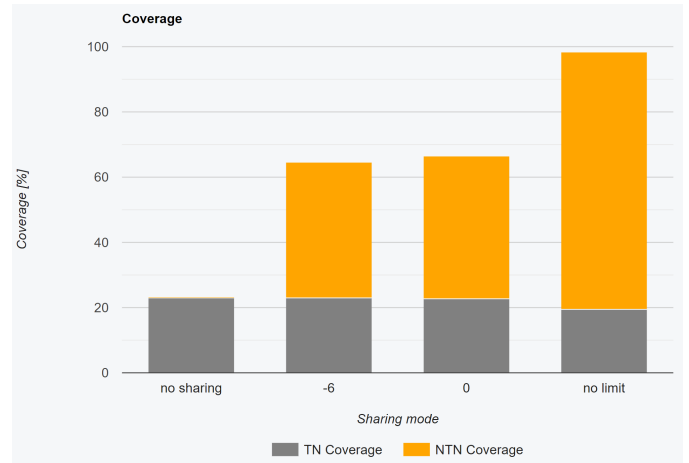


Figure 8: Coverage improvement in Centre-Val de Loire.

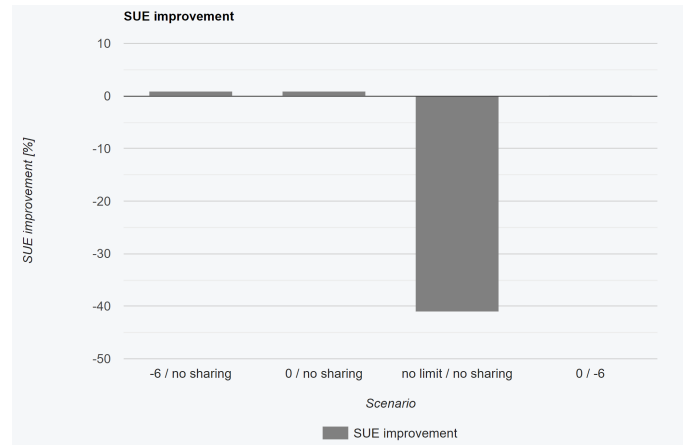


Figure 9: SUE improvement in Centre-Val de Loire.

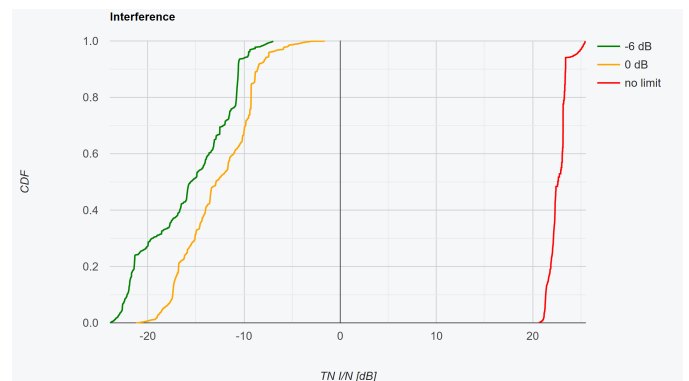


Figure 10: Interference in Centre-Val de Loire.

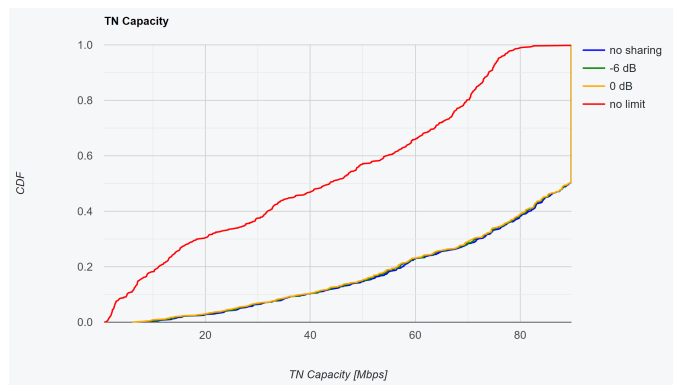


Figure 11: Capacity in Centre-Val de Loire.

site electricity and fibre connections. NTN can also be an option for backbone connection of community networks.

For the first time in the literature, a Proof of Concept is described for a DSA system, where LEO satellite beams can be activated and deactivated to protect TN deployments. With this study, we verified that spectrum sharing is a viable option when the TN and NTN coverage areas do not have to overlap. We verified that spectrum sharing improves significantly coverage and slightly increases the SUE. The spectrum management system is able to control the aggregate interference due to the spectrum sharing, and the impact of sharing in capacity of TN is extremely low. The possibilities for further developing the Proof of Concept include: aggregate interference analysis also in TN→NTN direction, more realistic study area having a land-sea border and a country border, evaluation of LEO movement, and the study of the number of UEs per TN basestation. In the next phase of the study, we will analyse how satellite movement impacts the spectrum sharing. We will research questions like how sharing results differ when satellites are moving towards nadir and away from nadir, and what is the performance degradation if the same optimization result is applied over a period of tens of seconds.

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